

1. EP & BVA

FR-01 – Validasi Durasi Pinjam

- Durasi peminjaman harus 1–4 jam
- Jika di luar rentang maka sistem menolak

1.1. Identifikasi Domain Input

Input	Domain
Duration	1–4 jam

1.2. Equivalence Class

- Valid Equivalence Class

Kelas	Range	Status
V1	$1 \leq \text{duration} \leq 4$	Valid

- Invalid Equivalence Class

Kelas	Range	Status
I1	$\text{duration} < 1$	Invalid
I2	$\text{duration} > 4$	Invalid

1.3. Test Case EP

Test Case	Input	Kelas	Expected Result
TC-EP-01	2	v1	Diterima
TC-EP-02	0	i1	Ditolak
TC-EP-03	7	i2	Ditolak

1.4. Boundary Value Analysis (BVA)

- Minimum Valid: 1
- Maximum Valid: 4
- Test Case Boundary Value: 0, 1, 2, 3, 4, 5
- Batas Minimum:

Value	Keterangan
0	Min -1
1	Minimum

2	Minimum +1
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- Batas Maximum:

Value	Keterangan
3	Maximum -1
4	Maximum
5	Maximum +1

1.5. Test Case BVA

Test Case	Input	Jenis Boundary	Expected Result
TC-BVA-01	0	Min-1	Ditolak
TC-BVA-02	1	Min	Diterima
TC-BVA-03	2	Min+1	Diterima
TC-BVA-04	3	Max-1	Diterima
TC-BVA-05	4	Max	Diterima
TC-BVA-06	5	Max+1	Ditolak

2. DECISION TABLE TESTING

FR-02 – Aturan Peminjaman Ruang

Mahasiswa boleh meminjam ruang jika:

- Mahasiswa terdaftar
- Tidak memiliki pelanggaran
- Ruang tersedia

2.1. Identifikasi Cause

- C1: Mahasiswa terdaftar
- C2: Tidak memiliki pelanggaran
- C3: Ruang tersedia

2.2. Identifikasi Effect

- E1: Booking berhasil
- E2: Ditolak (Mahasiswa tidak terdaftar)
- E3: Ditolak (Memiliki pelanggaran)

- E4: Ditolak (Ruang tidak tersedia)

2.3. Formula Logika

E1	C1 AND C2 AND C3
E2	NOT C1
E3	C1 AND NOT C2
E4	C1 AND C2 AND NOT C3

2.4. Enumerasi Kombinasi

Terdapat 3 cause:

$2^3 = 8$ kombinasi

2.5. Semua Kombinasi Input

Rule	C1	C2	C3
R1	T	T	T
R2	T	T	F
R3	T	F	T
R4	T	F	F
R5	F	T	T
R6	F	T	F
R7	F	F	T
R8	F	F	F

2.6. Mapping Kombinasi ke Effect

Rule	C1	C2	C3	Expected Result
R1	T	T	T	E1
R2	T	T	F	E4
R3	T	F	T	E3
R4	T	F	F	E3
R5	F	T	T	E2

R6	F	T	F	E2
R7	F	F	T	E2
R8	F	F	F	E2

2.7. Decision Table Sebelum Optimasi

Kondisi	R1	R2	R3	R4	R5	R6	R7	R8
C1	T	T	T	T	F	F	F	F
C2	T	T	F	F	T	T	F	F
C3	T	F	T	F	T	F	T	F
E1	T	F	F	F	F	F	F	F
E2	F	F	F	F	T	T	T	T
E3	F	F	T	T	F	F	F	F
E4	F	T	F	F	F	F	F	F

2.8. Optimasi dengan Don't Care

Kondisi	R1	R2	R3	R4
C1	True	True	True	False
C2	True	True	False	-
C3	True	False	-	-
Expected Result	Booking berhasil	Ruang tidak tersedia	Memiliki pelanggaran	Tidak terdaftar

Note:

"-" : don't care

2.9. Test Case Minimal

Test Case	Input	Expected Result
TC-DT-01	C1=T, C2=T, C3=T	Booking berhasil
TC-DT-02	C1=T, C2=T, C3=F	Ditolak: ruang tidak tersedia
TC-DT-03	C1=T, C2=F, C3=T	Ditolak: memiliki

		pelanggaran
TC-DT-04	C1=F, C2=T, C3=T	Ditolak: mahasiswa tidak terdaftar

3. STATE TRANSITION TESTING

FR-03 – Status Ruang

- Ruang hanya bisa dipinjam jika AVAILABLE
- Tidak boleh loncat state

3.1. Identifikasi State

Code	State
S1	AVAILABLE
S2	BOOKED
S3	IN_USE
S4	FINISHED

3.2. Identifikasi Event

Code	Event
E1	Booking
E2	Check-in
E3	Selesai
E4	Cancel

3.3. Initial dan Terminal State

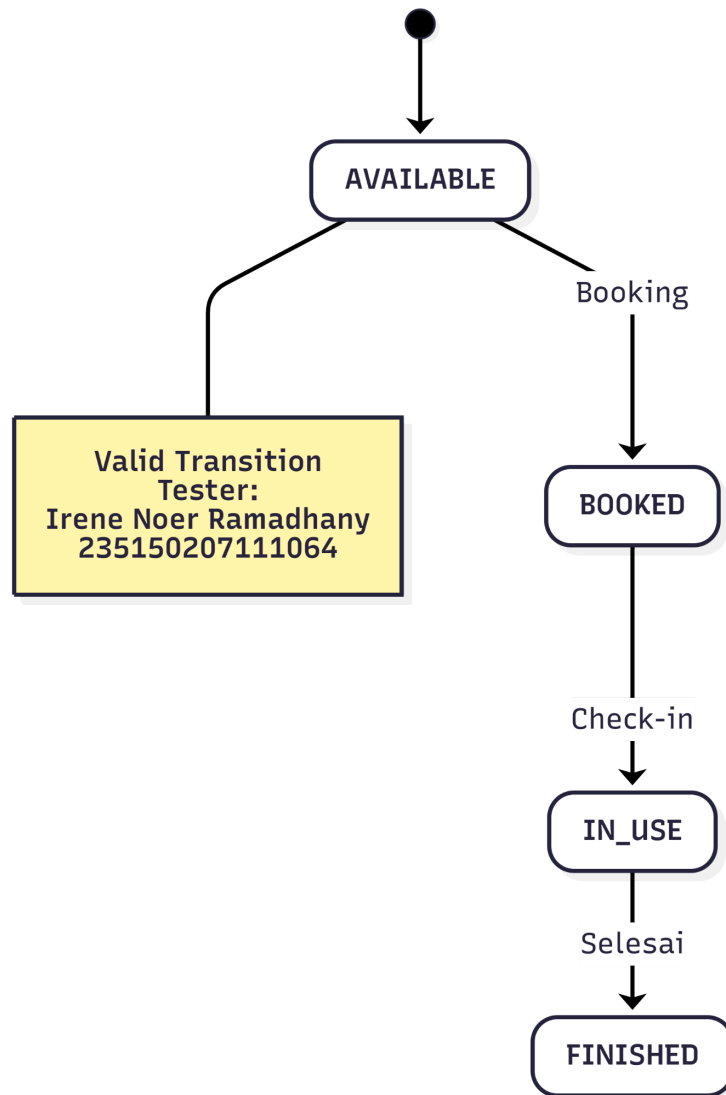
Jenis	State
Initial State	AVAILABLE
Terminal State	FINISHED

3.4. Transition Table

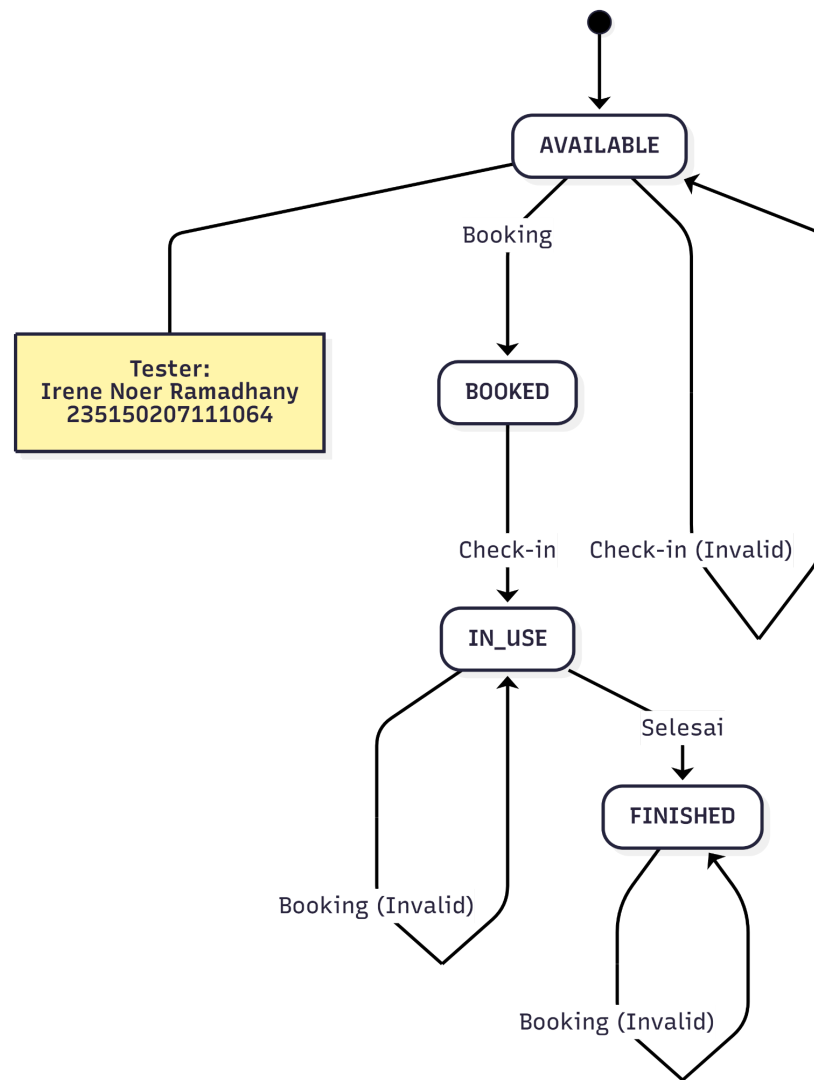
Current State	Event	Next State	Status
AVAILABLE	Booking	BOOKED	Valid

BOOKED	Check-in	IN_USE	Valid
IN_USE	Selesai	FINISHED	Valid
BOOKED	Cancel	AVAILABLE	Valid
AVAILABLE	Check-in	-	Invalid
AVAILABLE	Selesai	-	Invalid
IN_USE	Booking	-	Invalid
FINISHED	Booking	-	Invalid

3.5. Diagram Valid Transition



3.6. Diagram Invalid Transition



3.7. Test Case Valid Transition

Test Case	Current State	Event	Expected Result
TC-ST-01	AVAILABLE	Booking	BOOKED
TC-ST-02	BOOKED	Check-in	IN_USE
TC-ST-03	IN_USE	Selesai	FINISHED

3.8. Test Case Invalid Transition

Test Case	Current State	Event	Expected Result
TC-ST-04	AVAILABLE	Check-in	Ditolak

TC-ST-05	IN_USE	Booking	Ditolak
TC-ST-06	FINISHED	Booking	Ditolak

4. Implementasi Program

4.1. Kode Program

```

print("Tester: Irene Noer Ramadhany")
print("NIM    : 235150207111064")

print("\nTEST 1: EP & BVA TEST")

test_duration = [0,1,2,3,4,5]

for d in test_duration:
    if 1 <= d <= 4:
        print("Duration", d, "jam DITERIMA")
    else:
        print("Duration", d, "jam DITOLAK")

print("\nTEST 2: DECISION TABLE TEST")

def booking(mahasiswa, pelanggaran, ruang):
    if not mahasiswa:
        return "DITOLAK TIDAK TERDAFTAR"
    elif not pelanggaran:
        return "DITOLAK MEMILIKI PELANGGARAN"
    elif not ruang:
        return "DITOLAK RUANG TIDAK TERSEDIA"
    else:
        return "BOOKING BERHASIL"

test_cases = [
    (True, True, True),
    (False, True, True),
    (True, False, True),
    (True, True, False)
]

for tc in test_cases:
    print(tc, booking(*tc))

print("\nTEST 3: STATE TRANSITION TEST")

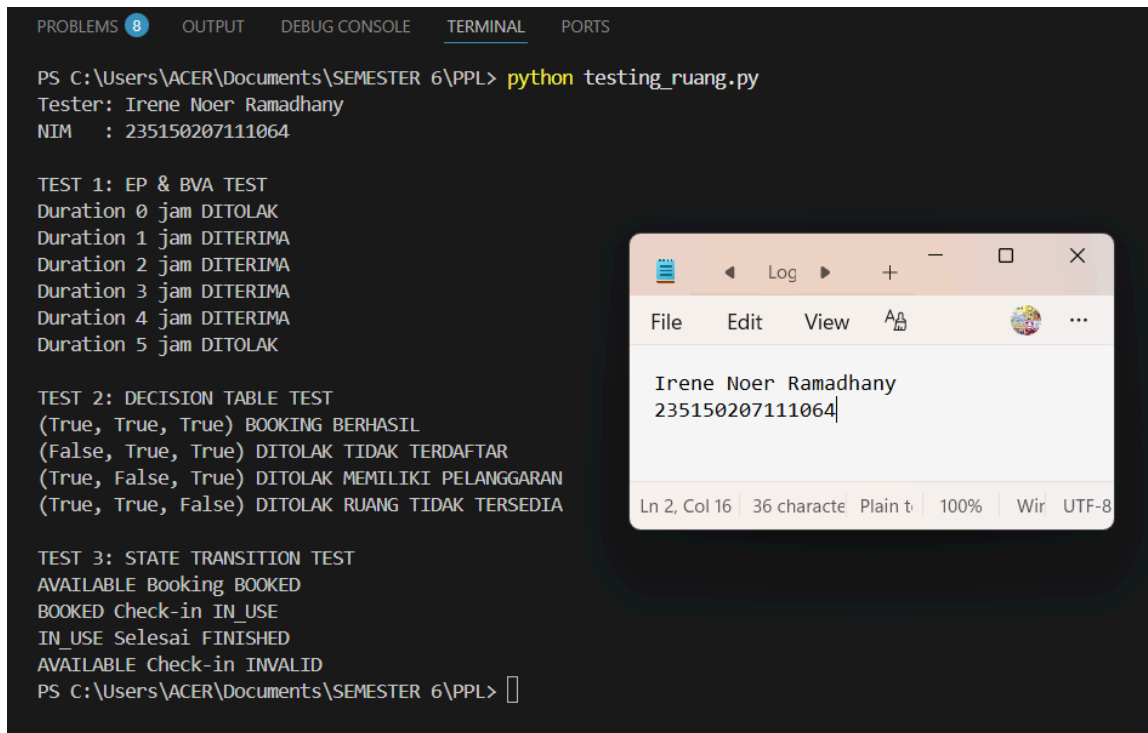
transitions = [
    ("AVAILABLE", "Booking", "BOOKED"),
    ("BOOKED", "Check-in", "IN_USE"),
    ("IN_USE", "Selesai", "FINISHED"),
    ("AVAILABLE", "Check-in", "INVALID"),
]

for t in transitions:

```

```
print(t[0], t[1], t[2])
```

4.2. Output



```
PS C:\Users\ACER\Documents\SEMESTER 6\PPL> python testing_ruang.py
Tester: Irene Noer Ramadhany
NIM : 235150207111064

TEST 1: EP & BVA TEST
Duration 0 jam DITOLAK
Duration 1 jam DITERIMA
Duration 2 jam DITERIMA
Duration 3 jam DITERIMA
Duration 4 jam DITERIMA
Duration 5 jam DITOLAK

TEST 2: DECISION TABLE TEST
(True, True, True) BOOKING BERHASIL
(False, True, True) DITOLAK TIDAK TERDAFTAR
(True, False, True) DITOLAK MEMILIKI PELANGGARAN
(True, True, False) DITOLAK RUANG TIDAK TERSEDIA

TEST 3: STATE TRANSITION TEST
AVAILABLE Booking BOOKED
BOOKED Check-in IN_USE
IN_USE Selesai FINISHED
AVAILABLE Check-in INVALID
PS C:\Users\ACER\Documents\SEMESTER 6\PPL>
```

5. LINK REPOSITORY

<https://github.com/irenenorr/blackbox-testing.git>

6. PERNYATAAN GEN AI

Saya menggunakan bantuan Gen-AI sekitar 40% untuk membantu memahami konsep Black-Box Testing, menyusun contoh test case, serta membantu struktur laporan. Seluruh hasil dari Gen-AI saya pelajari kembali, saya sesuaikan dengan requirement studi kasus, serta saya implementasikan dan modifikasi sendiri pada bagian kode program dan analisis pengujian.